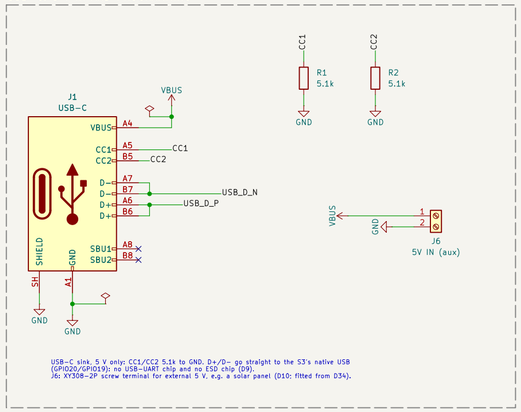
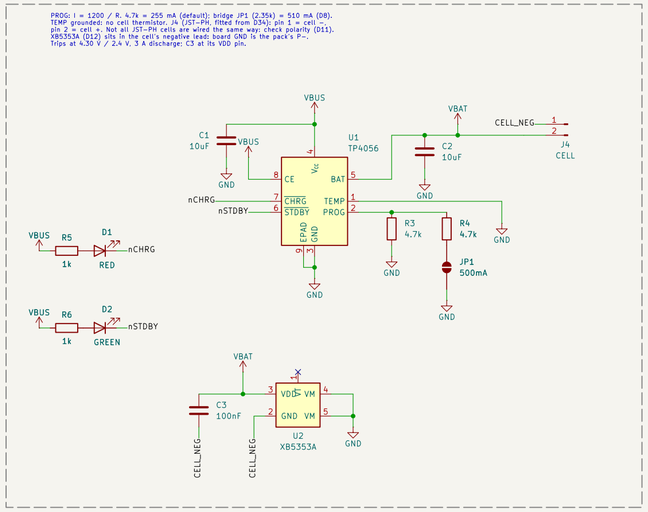
## USB-C: 5 V IN AND PROGRAMMING

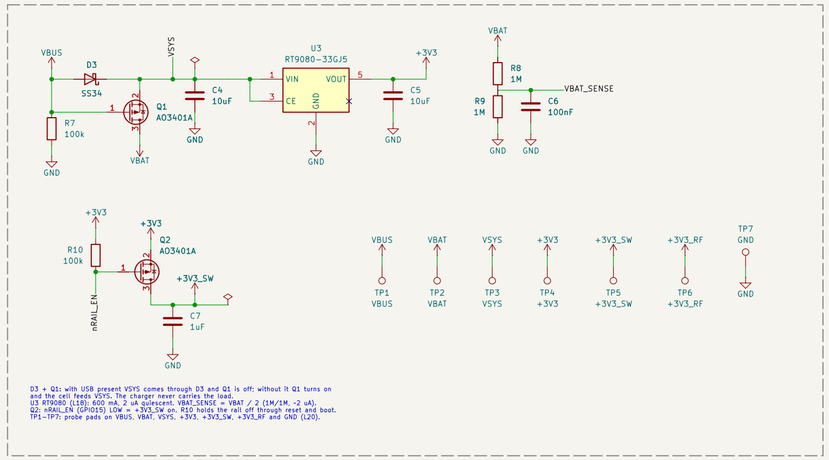



USB-C sink: 5 V only: CC1/CC2 5.1k to GND. D+/D− go straight to the S3's native USB (GPIO20/GPIO19); no USB-UART chip and no ESD chip (D9).
J6: XY308-2P screw terminal for external 5 V, e.g. a solar panel (D10; fitted from D34).

## LI-ION CHARGER & CELL PROTECTION

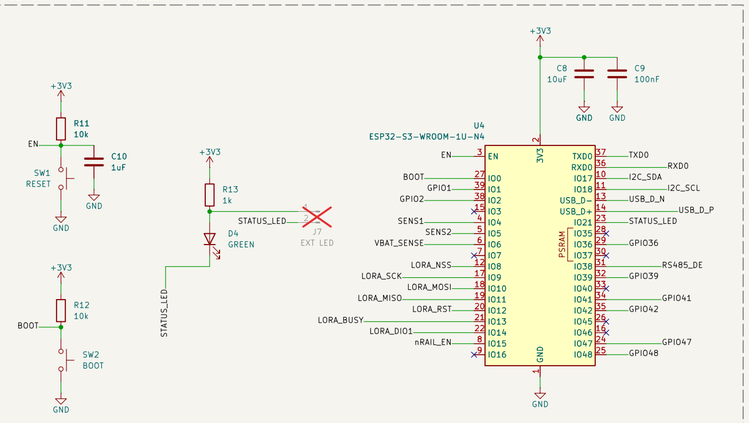



## SYSTEM POWER: LOAD SHARING, 3.3 V LDO, SWITCHED SENSOR RAIL



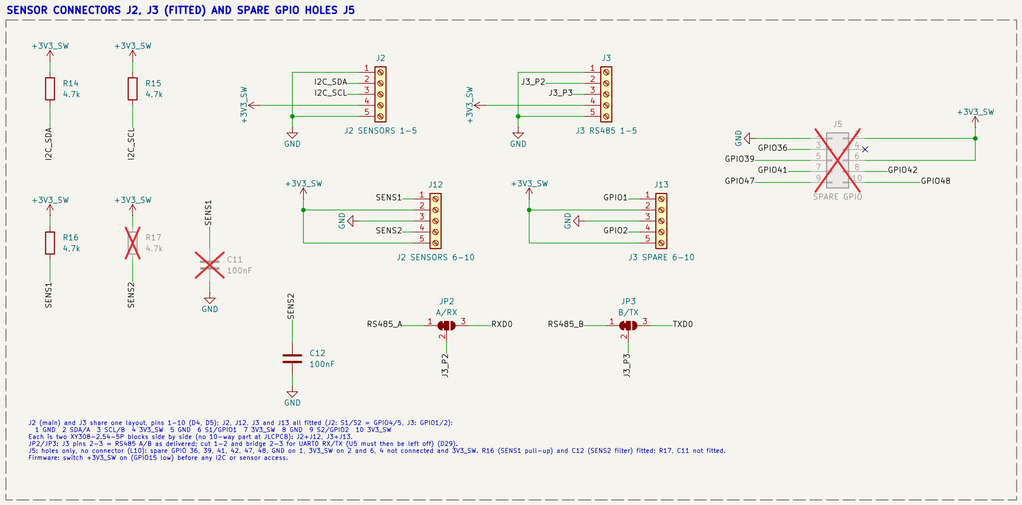

D3 + Q1: with USB present VSYS comes through D3 and Q1 is off; without it Q1 turns on and the cell feeds VSYS. The charger never carries the load.
U3 RT9080 (L18): 600 mA, 2 uA quiescent. VBAT_SENSE = VBAT // 2 (1M/1M, ~2 uA).
Q2: nRAIL_EN (GPIO15) LOW = +3V3_SW on. R10 holds the rail off through reset and boot.
TP1–TP7: probe pads on VBUS, VBAT, VSYS, +3V3, +3V3_SW, +3V3_RF and GND (L20).

## ESP32-S3-WROOM-1U MODULE



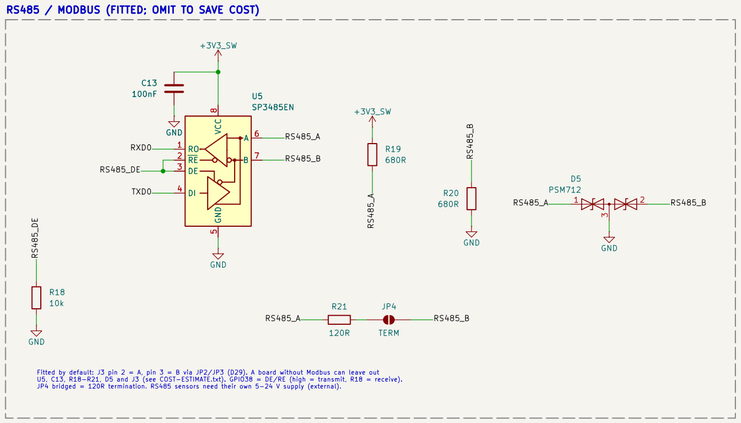

GPIO map (L3, L19): sensors 4/5 (ADC1) -> J2. spares 1/2 (ADC1) -> J13. VBAT_SENSE 6, I2C 17/18, LoRa NSS/SCK/MOSI/MISO 8–11, RST 12, BUSY 13, DIO1 14 (as Heltec V3/V4); nRAIL_EN 15, RS485 DE 36 (UART0 43/44 – RS485 or J3, USB 19/20, LED 21, BOOT 0.
Spares 36, 39, 41, 42, 47, 48 -> J5 holes (L10, L19). Not connected: straps 3, 45, 46 (internal defaults) and 7, 16, 35, 37, 40. J7 (not fitted): off-board LED, pin 1 = anode, pin 2 = cathode, shares R13. C8/C9 close to module pin 2; U.FL antenna: module end at the board edge, away from the LoRa antenna (L5).

## SENSOR CONNECTORS J2, J3 (FITTED) AND SPARE GPIO HOLES J5



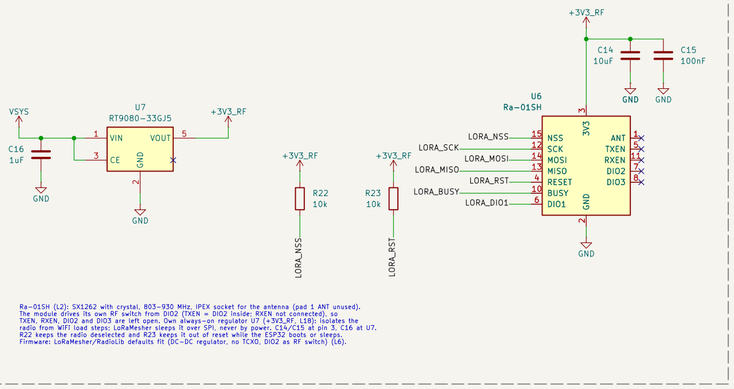

J2 (main) and J3 share one layout, pins 1–10 (D4, D5): J2, J12, J3 and J13 all fitted (J2: S1/S2 = GPIO4/5, J3: GPIO1/2):
1 GND 2 SDA/A 3 SCL/B 4 3V3_SW 5 GND 6 S1/GPIO1 7 3V3_SW 8 GND 9 S2/GPIO2 10 3V3_SW
Each is two XH308-2J4-5P blocks side by side (no 10-way part at JLCPCB): J2+J12, J3+J13.
JP2/JP3: J3 pins 2–3 = RS485 A/B as delivered; cut 1–2 and bridge 2–3 for UART0 RX/TX (U5 must then be left off) (D29).
J5: holes only, no connector (L10); spare GPIO 36, 39, 41, 42, 47, 48, GND on 1, 3V3_SW on 2 and 6, 4 not connected and 3V3_SW. R16 (SENS1 pull-up) and C12 (SENS2 filter) fitted: R17, C11 not fitted.
Firmware: switch +3V3_SW on (GPIO15 low) before any I2C or sensor access.

## RS485 / MODBUS (FITTED; OMIT TO SAVE COST)




Fitted by default: J3 pin 2 = A, pin 3 = B via JP2/JP3 (D29). A board without Modbus can leave out U5, C13, R18–R21, D5 and J3 (see COST-ESTIMATE.txt). GPIO38 = DE/RE (high = transmit, R18 = receive).
JP4 bridged = 120R termination. RS485 sensors need their own 5–24 V supply (external).

## LORA RADIO: AI-THINKER RA-01SH (SX1262)




Ra-01SH (L2): SX1262 with crystal, 803–930 MHz. IPEX socket for the antenna (pad 1 ANT unused).
The module drives its own RF switch from DIO2 (TXEN = DIO2 inside; RXEN not connected), so TXEN, RXEN, DIO2 and DIO3 are left open. Own always-on regulator U7 (+3V3_RF, L18): isolates the radio from WiFi load steps; LoRaMesher sleeps it over SPI, never by power. C14/C15 at pin 3, C16 at U7. R22 keeps the radio deselected and R23 keeps it out of reset while the ESP32 boots or sleeps.
Firmware: LoRaMesher/RadioLib defaults fit (DC-DC regulator, no TCXO, DIO2 as RF switch) (L6).

Generated by claude-tools/schematic/draw.py from circuit.py -- edit those, not this file

**Natural Innovation**

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File: frugal_iot_v03.kicad_sch

**Title: Frugal-IoT V0.3 LoRa**

| Size: A2 | Date: 2026-10-31 | **Rev: 0.3** |
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| KiCad E.D.A. 10.0.4 | | Id: 1/1 |